Lecture II: Explore & Exploit

Programming: Everyday Decision-Making Algorithms

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Today in one sentence

Whether to try something new or stick with what you know depends mostly on how much time you have left.

Why it matters

You have 45 minutes for lunch: the new restaurant or your reliable favorite? Companies face the same choice every day: a new supplier or the proven one, a new marketing channel or the one that works, a new film or another sequel.

Roadmap

1. The dilemma: try something new or stick with what works
2. Time decides: why the end of the interval changes everything
3. Smart strategies: from rules of thumb to the Gittins index

The dilemma: try something new or stick with what works

You have 45 minutes for lunch. Do you try the new restaurant or go to your reliable favorite?

📦 Quick Poll: Raise your hand, who would try the new restaurant?

Photo by Jillian Amatt on Unsplash

Explore gathers information, exploit uses it

🔍 **Explore**

- **Definition:** Gathering new information
- **Risk:** Unknown outcomes
- **Example:** Trying that new sushi place

🎯 **Exploit**

- **Definition:** Using known information
- **Risk:** Missing better options
- **Example:** Your go-to pizza joint

> ❗ Important
>
> In any given moment you can only do **one of the two**.

Examples from Your Life

Question: **Where do you face explore vs. exploit decisions?**

🏥 Medical: Clinical Trials

Explore: Test experimental drug on new patients

- *Risk:* Unknown side effects, potential harm
- *Reward:* Breakthrough treatment discovery

Exploit: Use proven treatment

- *Risk:* Missing better cures
- *Reward:* Predictable, safe outcomes

💻 Tech: A/B Testing

Explore: Test new website design

- *Cost:* Development time, potential lost conversions
- *Benefit:* Higher conversion rates, better UX

Exploit: Keep current design

- *Cost:* Missed optimization opportunities
- *Benefit:* Stable, known performance

💕 Personal: Dating Apps

Explore: Meet someone new

- *Risk:* Bad dates, wasted time
- *Reward:* Finding “the one”

Exploit: Meet someone you already know

- *Risk:* Settling for “good enough”
- *Reward:* Comfortable, known connection

The Daily Decision Dilemma

Explore Moments

- Try a new coffee shop
- Learn a new skill
- Watch a different genre
- Take a new route to work

Exploit Moments

- Order your usual drink
- Use familiar tools
- Rewatch favorite shows
- Take the fastest route

Question: **What is the optimal balance?**

Only exploiting: you miss better options

Question: Any ideas what it might be?

- Exploitation is not always the best strategy
- Especially with **very limited information**
- If you **stop exploring**, you **miss better options**

Only exploring: you never enjoy the results

Question: Any ideas what it might be?

- Exploration is not always the best strategy
- If you **constantly explore**, you **never enjoy the results**
- Imagine eating each meal, hearing each song, meeting each person **only once**

Computer scientists call it the multi-armed bandit

- Computer scientists formalized the explore-exploit trade-off
- It is known as the multi-armed bandit problem
- Question: What is a one-armed bandit?

One Armed Bandit

Photo by Kabir M on Unsplash

Three slot machines, 20 pulls: which do you play?

Imagine: You walk into a casino with **3 slot machines**

- Each machine has a **different** (unknown) win rate
- You have **$100** and **20 pulls** total
- **Goal:** Maximize your winnings
- **Explore:** try different machines. **Exploit:** play the best one so far

Every pull costs money and gives information

- Each pull tells us more about a machine's chance of winning
- But each pull also **costs** money
- Goal: maximize the total winnings

Question: **How do you decide which machine to play next?**

Interactive Casino Challenge

You've been playing for a while. Here's what you know:

Machine	Pulls	Wins	Win Rate	Your Choice?
A 🎰	15	9	60%	
B 🎰	2	1	50%	
C 🎰	0	0	???	

💡 Class Vote: A (known performer), B (limited data) or C (complete unknown)?

ℹ️ Follow-up: What if you only had **2 pulls** left? Or **100**?

Machine A has the higher average, but few pulls

- Expected value = average wins per pull
- E(A) = 9/15 = 0.6 and E(B) = 1/2 = 0.5
- But 2 and even 15 pulls are **few observations**

The best choice now is not always the best choice overall

- The bandit problem stands for many real-world decisions
- The best choice **now** can differ from the best choice **in the long run**
- Which lever to pull next depends on one more thing:

The dilemma: try something new or stick with what works

What we covered

Every choice between the new and the known trades information against reward.

Time decides: why the end of the interval changes everything

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How long will you stay in the casino?

- Question: Why does this matter?
- Question: How does it change our choice between machine A and B?

> "I'm more likely to try a new restaurant when I move to a city than when I'm leaving it" (Chris Stucchio)

Long interval: explore. Short interval: exploit.

- The time you plan to stay in the casino is **the interval**
- **Long interval:** explore more, you have time to use what you learn
- **Short interval:** exploit what you already know

Explore while you still have time to use what you learn

> "I moved to Pune, India, and I just [...] eat everywhere that didn't look like it was gonna kill me" (Chris Stucchio)

Exploration

Photo by Colin Maynard on Unsplash

Exploit when you are ready to cash in

> "And as I was leaving the city I went back to all my old favorites, rather than trying out new stuff [...]. Even if I find a slightly better place, I'm only going to go there once or twice, so why take the risk?" (Chris Stucchio)

Exploitation

Photo by Cristina Gottardi on Unsplash

Hollywood exploits: 9 of the top 10 films are sequels

ℹ Sequels among the 10 highest-grossing films

Year	Number of Sequels
1981	2
1991	3
2001	5
2011	8
2024	9[1]

Question: **What explains this trend, and what does it say about Hollywood's interval?**

Sequels…

The Insight: Studios are **exploiting** proven franchises because they think they're approaching the **"end of their interval"** (streaming disruption, changing audience preferences).

Photo by Universal Pictures on Wikipedia

Studios behave as if their interval is ending

- A new movie is risky, but can create a new fan base (**explore**)
- A sequel has a guaranteed fan base, a sure thing (**exploit**)
- Studios may believe they are near the **end of their interval**
- So they pull the best arms before the casino turns them out

Anecdotes are not a strategy

- Anecdotes help us understand the trade-off, but they are no solution
- An exact algorithm for the trade-off is **very hard** to find
- On the way, there were many interesting approaches…

Time decides: why the end of the interval changes everything

What we covered

The more time is left, the more exploring pays off; near the end, exploit what you know.

Smart strategies: from rules of thumb to the Gittins index

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Rule of thumb: win-stay, lose-shift[2]

- Question: What do you think this strategy does?
- **Won?** Stay with the same arm. **Lost?** Switch to another arm.

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[1]Box Office Mojo, top 10 by US domestic gross 2024 (boxofficemojo.com/year/2024). Counted as sequels: all except *Wicked*; *Twisters* counted as a sequel.

[2]For more details see Robbins, H. (1952) 'Some aspects of the sequential design of experiments', Bulletin of the American Mathematical Society, 58.

- Simple, and proven better than choosing an arm at random

Win-stay makes sense

- Question: What do you think about win-stay?
- If you win, pull the same arm again
- The arm has just become more attractive

Lose-shift is too hasty

- Question: What do you think about lose-shift?
- Switching after every loss is a rash move
- Your favorite restaurant, tenth visit: always great, today disappointing
- Should you never go back?

Easy answers come with problems

- Win-stay lose-shift **penalizes losses too much**
- It ignores **the interval**
- But it was a good start for better approaches

A fair estimate: Laplace's rule of succession

- First we need a fair estimate of each machine
- Laplace (18th century): after w wins and l losses, the chance of a win is about

$$\mathbb{E}[\text{win}] = \frac{w+1}{w+l+2}$$

- In words: add one imagined win and one imagined loss, then take the share of wins

Try it: machine A and B by Laplace

$$\mathbb{E}[\text{win}] = \frac{w+1}{w+l+2}$$

- Question: What is E[B]?
- Question: What is E[A]?
- Question: Which machine looks better by this estimate?

Laplace estimates for wins and losses

Losses/Wins	1	2	3	4	5	6	7	8	9	10
1	0.50	0.60	0.67	0.71	0.75	0.78	0.80	0.82	0.83	0.85
2	0.40	0.50	0.57	0.63	0.67	0.70	0.73	0.75	0.77	0.79
3	0.33	0.43	0.50	0.56	0.60	0.64	0.67	0.69	0.71	0.73
4	0.29	0.38	0.44	0.50	0.55	0.58	0.62	0.64	0.67	0.69

Losses/Wins	1	2	3	4	5	6	7	8	9	10
5	0.25	0.33	0.40	0.45	0.50	0.54	0.57	0.60	0.63	0.65
6	0.22	0.30	0.36	0.42	0.46	0.50	0.53	0.56	0.59	0.61
7	0.20	0.27	0.33	0.38	0.43	0.47	0.50	0.53	0.56	0.58
8	0.18	0.25	0.31	0.36	0.40	0.44	0.47	0.50	0.53	0.55
9	0.17	0.23	0.29	0.33	0.38	0.41	0.44	0.47	0.50	0.52
10	0.15	0.21	0.27	0.31	0.35	0.39	0.42	0.45	0.48	0.50

Bellman: calculate backwards from the last pull

- Richard Bellman found the exact solution for a **known interval**
- Start at the **final pull**: with no future left, simply exploit
- Then calculate **backwards** the best choice in every earlier situation

Bellman needs to know how long you stay

- The calculation grows fast with many arms and a long interval
- Worse: we usually **don't know** how long we stay in the casino
- So the bandit problem seemed to have no practical solution

Gittins: a drug-trial problem at Unilever[3]

- 1970s: Unilever asks the mathematician John Gittins to optimize its drug trials
- Which compound is effective, found out as quickly as possible?
- Gittins generalized the problem and found its **optimal solution**

Gittins discounts the future instead of fixing the interval

- Earlier solutions needed a **known interval**
- Gittins instead **discounts** future wins, like future cash flows
- So any interval length works, even an endless one

Discounting: tomorrow counts less than today

- Future rewards count less than immediate ones (**discount factor**)
- **Money:** Money today can earn interest
- **Everyday life:** Tonight's dinner beats the same dinner next week

Back to machine A and B

- The Gittins index works for every multi-armed bandit problem
- Let's consider our two machines one last time:
  - Machine A: 15 pulls, 9 wins, 6 losses

[3]For more details see Gittins, J. (1979) 'Bandit Processes and Dynamic Allocation Indices', Journal of the Royal Statistical Society. Series B (Methodological).

- Machine B: 2 pulls, 1 win, 1 loss

The Gittins index as a lookup table

Gittins index for different numbers of wins and losses (discount factor 0.9).

Losses/Wins	0	1	2	3	4	5	6	7	8	9
0	.7029	.8001	.8452	.8723	.8905	.9039	.9141	.9221	.9287	.9342
1	.5001	.6346	.7072	.7539	.7869	.8115	.8307	.8461	.8588	.8695
2	.3796	.5163	.6010	.6579	.6996	.7318	.7573	.7782	.7956	.8103
3	.3021	.4342	.5184	.5809	.6276	.6642	.6940	.7187	.7396	.7573
4	.2488	.3720	.4561	.5179	.5676	.6071	.6395	.6666	.6899	.7101
5	.2103	.3245	.4058	.4677	.5168	.5581	.5923	.6212	.6461	.6677
6	.1815	.2871	.3647	.4257	.4748	.5156	.5510	.5811	.6071	.6300
7	.1591	.2569	.3308	.3900	.4387	.4795	.5144	.5454	.5723	.5960
8	.1413	.2323	.3025	.3595	.4073	.4479	.4828	.5134	.5409	.5652
9	.1269	.2116	.2784	.3332	.3799	.4200	.4548	.4853	.5125	.5373

By the Gittins index, machine B wins

Choose machine A or B according to the Gittins index?

- Machine B (0.6346) beats machine A (0.6300)
- Clear **win-stay** pattern, relaxed **lose-shift** pattern
- At (0,0) we see the **exploration bonus**

Gittins Index: The Scary Math

$$G_i(s_i, f_i) := \sup_{\tau \geq 1} \frac{\mathbb{E}\left[\sum_{t=0}^{\tau-1} \beta^t \cdot r_i^t \;\middle|\; s_i, f_i\right]}{\mathbb{E}\left[\sum_{t=0}^{\tau-1} \beta^t\right]}$$

In words: the best average reward you can expect from this arm if you may stop playing it whenever you like.

> ! Important
>
> **Too complex to calculate by hand**, and you don't need to: just use the lookup table!

Higher index, better choice

- **Core idea:** "What's the best possible average reward from this option?"
- **Higher index = better choice**
- Less-tested options get an **exploration bonus**, that's why machine B wins

- The index balances exploration and exploitation automatically

Which rule when?

- Both outcomes known: choose the higher expected value
- E(explore) unknown, interval known: **Bellman** (backward calculation)
- E(explore) unknown, interval unknown: **Gittins index**

🎯 Your Personal Action Plan

Question: **Rate yourself from 1 (exploit) to 10 (explore):**

- **Career:** new skills and roles vs. deepening your expertise
- **Social life:** meeting new people vs. investing in friendships
- **Learning:** new subjects vs. mastering current interests

Time horizon matters: more time left, more exploring.

Smart strategies: from rules of thumb to the Gittins index

What we covered

Simple rules like win-stay, lose-shift help, but better strategies weigh both what you know and how uncertain you still are.

What we covered

The dilemma: try something new or stick with what works

Every choice between the new and the known trades information against reward.

Time decides: why the end of the interval changes everything

The more time is left, the more exploring pays off; near the end, exploit what you know.

Smart strategies: from rules of thumb to the Gittins index

Simple rules like win-stay, lose-shift help, but better strategies weigh both what you know and how uncertain you still are.

Why it matters

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Explore early.

Exploit late.

Literature

Interesting literature to start

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.[4]

Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Free online
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

> **i** Note
>
> Think Python is free online and a great start. Schrödinger programmiert Python is a playful alternative in German.

More Literature

For more interesting literature, take a look at the literature list of this course.

[4] The main inspiration for this lecture. We have read it and discussed it in depth, always wanting to translate it into a course.